

Volume I. First Section.
§§ 5–8. Rational functions.

§ 5. Rational fractional functions; divisibility.

If $F(x)$ and $f(x)$ are two integral rational functions of x , then the fraction

$$\frac{F(x)}{f(x)} \quad (1)$$

is called a *fractional rational*, or also briefly a *fractional* or *rational function* of x .

If the degree of the numerator is lower than the degree of the denominator, then the function is called a *proper fraction*; in the opposite case it is called an *improper fraction*.

According to § 3 the integral rational functions Q and $\varphi(x)$ can be so determined that

$$F(x) = Qf(x) + \varphi(x) \quad (2)$$

and the degree of $\varphi(x)$ is smaller than the degree of $f(x)$; consequently

$$\frac{F(x)}{f(x)} = Q + \frac{\varphi(x)}{f(x)}, \quad (3)$$

and hence the theorem:

Every fractional function can be decomposed into the sum of an integral function and a proper fractional function.

If n is the degree of $f(x)$, then the degree of $\varphi(x)$ is at most $n - 1$; it may also be lower. In particular, the case may occur in which $\varphi(x)$ vanishes identically.

In this case the function $F(x)$ is said to be divisible by $f(x)$. The function

$$\frac{F(x)}{f(x)}$$

is in this case only apparently fractional; in reality it is equal to the integral function Q .

The formula

$$\frac{x^m - 1}{x - 1} = x^{m-1} + x^{m-2} + x^{m-3} + \cdots + 1$$

gives a simple example of this.

For divisibility of functions the same laws hold as for divisibility of numbers, in particular the following:

1. If the function $F(x)$ is divisible by the function $f(x)$, and $f(x)$ is divisible by a third function $\varphi(x)$, then $F(x)$ is also divisible by $\varphi(x)$.

For if

$$F = Qf, \quad f = q\varphi,$$

where Q and q are integral rational functions, then

$$F = Qq\varphi,$$

and since Qq is an integral rational function, F is divisible by φ .

2. If $F(x)$ is divisible by $f(x)$ and Q is any integral rational function, then $QF(x)$ is also divisible by $f(x)$.

3. If $F(x)$ and $f(x)$ are divisible by $\varphi(x)$, then $F(x) \pm f(x)$ is also divisible by $\varphi(x)$, or more generally:

4. If $F_1, F_2 \dots$ are divisible by $f(x)$ and $Q_1, Q_2 \dots$ are arbitrary integral rational functions, then

$$Q_1 F_1 + Q_2 F_2 + \dots$$

is also divisible by $f(x)$.

The last theorem includes the two preceding ones and is proved simply as follows.

If $F_1, F_2 \dots$ are divisible by f , then the integral rational functions $\Phi_1, \Phi_2 \dots$ can be determined so that

$$F_1 = \Phi_1 f, \quad F_2 = \Phi_2 f, \dots$$

and consequently

$$Q_1 F_1 + Q_2 F_2 + \dots = (Q_1 \Phi_1 + Q_2 \Phi_2 + \dots) f.$$

Since $Q_1 \Phi_1 + Q_2 \Phi_2 + \dots$ is an integral rational function, the theorem is proved.

5. Every function is divisible by itself.

With respect to divisibility or indivisibility of functions nothing is changed if the functions are multiplied by arbitrary factors independent of x .

A non-zero quantity independent of x may be regarded as a function of degree zero. If we call such a quantity a constant, we may say:

6. Every function is divisible by every constant.

If a function is divisible by another, then the degree of the quotient is equal to the difference between the degree of the dividend and the degree of the divisor. The former therefore cannot be smaller than the latter. If the degrees are equal, the quotient is a constant, and hence:

7. If, of two integral rational functions of the same degree, one is divisible by the other, then they differ from one another only by a constant factor, and the second is also divisible by the first.

8. By § 4, the necessary and sufficient condition that the function $f(x)$ be divisible by the linear function $x - \alpha$ is that $f(\alpha) = 0$.

§ 6. Greatest common divisor.

It is a problem of fundamental importance to decide whether two integral rational functions have, besides the constants, any further common divisor. The solution is found by the algorithm of the greatest common divisor in exactly the same way as the corresponding question concerning divisibility of integers is answered. See the Introduction.

Let

$$f(x) = A, \quad \varphi(x) = A' \tag{1}$$

be two given functions; the degree of $\varphi(x)$ is supposed to be lower, or at least not higher, than that of $f(x)$.

We divide A by A' and denote the remainder, whose degree is lower than the degree of A' , by A'' , thus:

$$A = Q' A' + A'';$$

then we divide A' by A'' and denote the remainder by A''' , and continue in this way in forming the sequence of functions

$$A, A', A'', A''', \dots, \tag{2}$$

whose degrees $n, n', n'', n''' \dots$ always decrease and therefore, after a finite number of divisions, necessarily go down to zero. Let the last remainder of degree zero, and therefore a constant, be

$A^{(\nu)}$. Then we have the chain of equations

$$\begin{aligned} A &= Q'A' + A'', \\ A' &= Q''A'' + A''', \\ &\dots \\ A^{(\nu-3)} &= Q^{(\nu-2)}A^{(\nu-2)} + A^{(\nu-1)}, \\ A^{(\nu-2)} &= Q^{(\nu-1)}A^{(\nu-1)} + A^{(\nu)}, \end{aligned} \tag{3}$$

where $A^{(\nu)}$ is a constant.

If now A, A' have any common divisor, then by the theorems of the preceding paragraph, as is seen from the equations (3), it is also a divisor of A'', A''', A'''' and so on down to $A^{(\nu)}$. If $A^{(\nu)}$ is a non-zero constant, then no divisor of $f(x)$ and $\varphi(x)$ depending on x can exist. Such functions are called *prime to one another*, or *relatively prime*. One also says, taking account only of divisors depending on x , that the functions have no common divisor.

The condition that the two functions have a common divisor is therefore

$$A^{(\nu)} = 0. \tag{4}$$

In this case $A^{(\nu-1)}$ is a divisor of $A^{(\nu-2)}$, as the last equation (3) shows, and by the preceding equation it is a divisor of $A^{(\nu-3)}$, and so on, hence also a divisor of A and A' , that is, of f and φ . Conversely, every common divisor of A, A' is also a divisor of $A^{(\nu-1)}$; accordingly $A^{(\nu-1)}$ is called the *greatest common divisor* of f and φ . The algorithm (3) shows that $A^{(\nu)}$ or $A^{(\nu-1)}$ can be derived from the coefficients of f and φ by the rational operations of arithmetic, and in such a manner that division is always only by the coefficients of the highest powers of x in the functions $A, A', A'', \dots$ which are not zero.

We shall apply these considerations to two examples.

First let

$$\begin{aligned} A &= a_0x^2 + a_1x + a_2, \\ B &= b_0x^2 + b_1x + b_2 \end{aligned} \tag{5}$$

be two functions of the second degree, so that a_0, b_0 are non-zero.

The first step is to form the equation

$$A = \frac{a_0}{b_0}B + C, \tag{6}$$

where

$$C = c_0x + c_1 \tag{7}$$

and

$$c_0 = \frac{a_1b_0 - b_1a_0}{b_0}, \quad c_1 = \frac{a_2b_0 - b_2a_0}{b_0}. \tag{8}$$

If $c_0 = 0$, so that C is constant, then A, B have a common factor only if also $c_1 = 0$; and then A is divisible by B , that is, A and B differ only by a constant factor. If however c_0 is different from zero, we continue the calculation by putting

$$B = QC + D,$$

where, according to § 4, if there $\alpha = -c_1 : c_0$ is put,

$$D = \frac{b_0c_1^2 - b_1c_0c_1 + b_2c_0^2}{c_0^2}. \tag{9}$$

If D is non-zero, then A, B have no common divisor. If however $D = 0$, then C is the greatest common factor of A and B .

Substituting the values c_0, c_1 from (8) into (9), and suppressing the denominator $b_0 c_0^2$, one obtains the condition for the existence of a common divisor C of A and B in the form

$$\begin{aligned} a_0^2 b_2^2 + a_2^2 b_0^2 - 2a_0 a_2 b_0 b_2 - a_1 a_2 b_0 b_1 - a_0 a_1 b_1 b_2 \\ + a_0 a_2 b_1^2 + a_1^2 b_0 b_2 = 0 \end{aligned} \quad (10)$$

or

$$(a_0 b_2 - b_0 a_2)^2 + (a_0 b_1 - a_1 b_0)(a_2 b_1 - a_1 b_2) = 0. \quad (11)$$

This condition is also fulfilled when c_0 and c_1 are both zero, and is therefore the necessary and sufficient condition that A, B have a common divisor.

The left hand side of (10) or (11) is called the *resultant* of the functions A and B .

As a second example we take the one already chosen in § 3. Put

$$\begin{aligned} f(x) &= a_0 x^3 + a_1 x^2 + a_2 x + a_3 = A, \\ f'(x) &= 3a_0 x^2 + 2a_1 x + a_2 = B, \end{aligned} \quad (12)$$

and suppose a_0 different from zero. Then by § 3 we have

$$A = QB + C, \quad (13)$$

$$C = c_0 x + c_1, \quad (14)$$

where

$$c_0 = \frac{6a_0 a_2 - 2a_1^2}{9a_0}, \quad c_1 = \frac{9a_0 a_3 - a_1 a_2}{9a_0}. \quad (15)$$

If c_0 is zero, the algorithm is already closed. If c_1 is non-zero, then A and B have no common divisor; but if c_1 is also zero, then B itself is the greatest common divisor of A and B , that is, A is divisible by B . The conditions for this are therefore

$$3a_0 a_2 - a_1^2 = 0, \quad 9a_0 a_3 - a_1 a_2 = 0. \quad (16)$$

If c_0 is not zero, we go one step further and put

$$B = PC + D, \quad (17)$$

where D becomes constant and has the expression

$$D = \frac{a_2 c_0^2 - 2a_1 c_0 c_1 + 3a_0 c_1^2}{c_0^2}. \quad (18)$$

If this expression is non-zero, then A and B are prime to one another; if it is zero, then A and B have the greatest common divisor C . If for c_0, c_1 one substitutes the values (15), omits the denominator, still removes the non-vanishing factor $9a_0$, and changes the sign, this condition, after a simple calculation, assumes the form

$$a_1^2 a_2^2 + 18a_0 a_1 a_2 a_3 - 4a_0 a_2^3 - 4a_1^3 a_3 - 27a_0^2 a_3^2 = 0. \quad (19)$$

It is, as one readily sees either by calculation or from (18), also fulfilled when the conditions (16) hold, and is therefore the necessary and sufficient condition that $f(x)$ and $f'(x)$ have a common divisor. The left hand side of (19), which is an integral rational and homogeneous function of the coefficients of $f(x)$, is called the *discriminant* of the function $f(x)$.

We derive one more theorem from the algorithm (3), which is often used.

From the first of these equations it follows that

$$A'' = A - Q'A', \quad (20)$$

and if this value of A'' is substituted in the following equation, one obtains

$$A''' = (1 + Q'Q'')A' - Q''A,$$

thus, denoting by p, p' integral rational functions,

$$A''' = pA + p'A'. \quad (21)$$

Substituting the expressions (20), (21) in the third equation (3), one obtains for A''' again an expression of the form (21), and so one may continue, obtaining finally

$$A^{(\nu)} = PA + P'A', \quad (22)$$

where P, P' are integral rational functions whose coefficients are composed from the coefficients of A and A' by rational operations.

In the formula (22), $A^{(\nu)}$ is a constant. This theorem is especially important in the case where $A^{(\nu)}$ is non-zero, so that A, A' are relatively prime. If in this case we put

$$P = A^{(\nu)}F(x), \quad P' = A^{(\nu)}\Phi(x),$$

then, after suppressing the factor $A^{(\nu)}$, the theorem can be expressed as follows:

I. If $f(x)$ and $\psi(x)$ are two integral functions without a common divisor, then two other integral functions $F(x)$ and $\Phi(x)$ can be determined which satisfy identically the equation

$$F(x)f(x) + \Phi(x)\psi(x) = 1. \quad (23)$$

The theorem can still be generalized. Multiplying (23) by an arbitrary integral function $\chi(x)$ gives

$$F(x)\chi(x)f(x) + \Phi(x)\chi(x)\psi(x) = \chi(x), \quad (24)$$

and by § 3 we can put

$$\Phi(x)\chi(x) = Q(x)f(x) + \varphi(x) \quad (25)$$

in such a way that $Q(x), \varphi(x)$ are integral rational functions of x and the degree of $\varphi(x)$ is lower than that of $f(x)$. Substituting this in (24), and putting in place of $F(x)\chi(x) + Q(x)\psi(x)$ again $F(x)$, we obtain

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x).$$

We can therefore generalize the preceding theorem as follows:

II. If $f(x), \psi(x), \chi(x)$ are given integral rational functions, and $f(x)$ and $\psi(x)$ are without a common divisor, then the integral rational functions $F(x), \varphi(x)$ can be so determined - with $\varphi(x)$ of lower degree than $f(x)$ - that the equation

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x) \quad (26)$$

is identically satisfied.

§ 7. Products of linear factors.

According to § 1, by multiplication of integral functions we again obtain such functions of higher degree, and the degree of the product is determined by the sum of the degrees of the individual factors.

If therefore we multiply n linear factors by one another, there arises a function of the n th degree, whose manner of formation we must examine somewhat more closely.

We shall take the linear factors in the simple form $x - \alpha_1, x - \alpha_2, \dots, x - \alpha_n$, and put, since the coefficient of the highest power is, by § 1, equal to 1,

$$\begin{aligned} f(x) &= (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \\ &= x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n. \end{aligned} \quad (1)$$

A slight consideration shows the following law of formation of the coefficients $a_1, a_2 \dots a_n$:

$-a_1$ is equal to the sum of the α 's; a_2 is equal to the sum of the products of every two of the α 's; $-a_3$ is the sum of the products of every three of the α 's; in general $(-1)^\nu a_\nu$ is the sum of the products of every ν of the quantities α , or, in formulas,

$$\begin{aligned} -a_1 &= \sum \alpha_1, \\ +a_2 &= \sum \alpha_1 \alpha_2, \\ &\dots \\ (-1)^\nu a_\nu &= \sum \alpha_1 \alpha_2 \cdots \alpha_\nu, \\ &\dots \\ (-1)^n a_n &= \alpha_1 \alpha_2 \cdots \alpha_n. \end{aligned} \quad (2)$$

But to see still more clearly the correctness of this law of formation, one uses complete induction.

One first verifies the correctness in the first cases by actually carrying out the multiplication:

$$\begin{aligned} (x - \alpha_1)(x - \alpha_2) &= x^2 - (\alpha_1 + \alpha_2)x + \alpha_1 \alpha_2, \\ (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) &= x^3 - (\alpha_1 + \alpha_2 + \alpha_3)x^2 \\ &\quad + (\alpha_1 \alpha_2 + \alpha_1 \alpha_3 + \alpha_2 \alpha_3)x - \alpha_1 \alpha_2 \alpha_3. \end{aligned}$$

If we assume the law of formation proved for $n - 1$ factors, and put

$$(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_{n-1}) = x^{n-1} + a'_1 x^{n-2} + \cdots + a'_{n-1},$$

then multiplication by $x - \alpha_n$ gives

$$\begin{aligned} a_1 &= a'_1 - \alpha_n, \\ a_2 &= a'_2 - \alpha_n a'_1, \\ a_3 &= a'_3 - \alpha_n a'_2, \\ &\dots \\ a_\nu &= a'_\nu - \alpha_n a'_{\nu-1}, \\ &\dots \\ a_n &= -\alpha_n a'_{n-1}, \end{aligned} \quad (3)$$

and from this the correctness of the formulas (2) is immediately evident.

It is important to determine the number of terms which occur in each of the sums (2). We denote the number of terms occurring in the sum $(-1)^\nu a_\nu$ by $B_\nu^{(n)}$; it is the number of combinations without repetition of n elements to the ν th class, that is, of every ν distinct ones among the n elements. To determine it, suppose first that the $B_{\nu-1}^{(n)}$ combinations to the $(\nu - 1)$ st class have been formed. From each of these combinations, by adding one at a time of the missing $n - \nu + 1$ elements, one can derive $n - \nu + 1$ combinations to the ν th class. In this way however every combination is formed ν times, namely by addition of each of its elements; hence one obtains the relation

$$B_\nu^{(n)} = B_{\nu-1}^{(n)} \frac{n - \nu + 1}{\nu}, \quad (4)$$

whereas $B_1^{(n)}$ plainly has the value n . Therefore, if one multiplies the equations following from (4),

$$\begin{aligned} B_1^{(n)} &= n, \\ B_2^{(n)} &= B_1^{(n)} \frac{n-1}{2}, \\ &\dots \\ B_\nu^{(n)} &= B_{\nu-1}^{(n)} \frac{n-\nu+1}{\nu}, \end{aligned} \tag{5}$$

it follows that

$$B_\nu^{(n)} = \frac{n(n-1)(n-2)\cdots(n-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu}. \tag{6}$$

In what follows we shall often, when m is an arbitrary positive integer, use the symbol

$$\Pi(m) = 1 \cdot 2 \cdot 3 \cdots m, \quad \Pi(0) = 1, \tag{7}$$

so that

$$\Pi(m) = m\Pi(m-1). \tag{8}$$

With the help of this symbol the expression for $B_\nu^{(n)}$ can be displayed more conveniently as

$$B_\nu^{(n)} = B_{n-\nu}^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)}, \tag{9}$$

which, if $B_0^{(n)} = 1$ is put, is also valid for $\nu = 0$ and $\nu = n$, and makes evident the invariance of $B_\nu^{(n)}$ under the interchange of ν with $n - \nu$.

The derivation of the formula (4), which we have just given according to the rules of the theory of combinations, is indeed perfectly correct and clear, but for its exact foundation it requires some reflection which cannot easily be compressed into a few words. We shall therefore subsequently prove its correctness once more by the method of complete induction.

The formulas (3), namely, give the following recurrence formulas:

$$\begin{aligned} B_1^{(n)} &= B_1^{(n-1)} + 1, \\ B_2^{(n)} &= B_2^{(n-1)} + B_1^{(n-1)}, \\ &\dots \\ B_\nu^{(n)} &= B_\nu^{(n-1)} + B_{\nu-1}^{(n-1)}, \\ &\dots \\ B_n^{(n)} &= B_{n-1}^{(n-1)}. \end{aligned} \tag{10}$$

Now the formula (6) can be confirmed very easily by counting for the first values of n . If one assumes it correct for $n - 1$, then (10) gives

$$B_\nu^{(n)} = \frac{\Pi(n-1)}{\Pi(\nu)\Pi(n-\nu-1)} + \frac{\Pi(n-1)}{\Pi(\nu-1)\Pi(n-\nu)},$$

whence, by (8),

$$B_\nu^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)},$$

and by this the general validity of formula (9) is proved.

From the meaning of $B_\nu^{(n)}$ it follows, and it is also proved from the formulas (10) by complete induction, that the $B_\nu^{(n)}$ are positive integers.

§ 8. The binomial theorem.

If in formula (1) of the preceding paragraph we set the previously arbitrary quantities $\alpha_1, \alpha_2 \dots \alpha_n$ equal to one another, we obtain the binomial theorem, which consists in nothing other than arranging the n th power of a binomial $x + y$ according to powers of x . Namely, if we put

$$\alpha_1 = \alpha_2 = \dots = \alpha_n = -y,$$

then formula (2) gives

$$a_\nu = (-1)^\nu \sum \alpha_1 \alpha_2 \dots \alpha_\nu = y^\nu B_\nu^{(n)},$$

where, according to the definition of the preceding paragraph, $B_\nu^{(n)}$ means the number of terms of the sum, which all become equal to one another and equal to $(-1)^\nu y^\nu$.

Accordingly one obtains

$$(x + y)^n = x^n + B_1^{(n)} x^{n-1} y + B_2^{(n)} x^{n-2} y^2 + \dots + B_n^{(n)} y^n, \quad (1)$$

or, written out,

$$\begin{aligned} (x + y)^n &= x^n + nx^{n-1}y + \frac{n(n-1)}{1 \cdot 2} x^{n-2}y^2 + \dots + y^n \\ &= \Pi(n) \sum \frac{x^{n-\nu} y^\nu}{\Pi(n-\nu)\Pi(\nu)} = \Pi(n) \sum \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \end{aligned}$$

the last sum being extended over all non-negative integral values α, β which satisfy the condition $\alpha + \beta = n$.

Accordingly the coefficients $B_\nu^{(n)}$ are called the binomial coefficients.

For reference we put here a small table of the first values of the binomial coefficients:

$n = 1$	1	1							
$n = 2$	1	2	1						
$n = 3$	1	3	3	1					
$n = 4$	1	4	6	4	1				
$n = 5$	1	5	10	10	5	1			
$n = 6$	1	6	15	20	15	6	1		
$n = 7$	1	7	21	35	35	21	7	1	

Among the various properties of the binomial coefficients we shall derive two, of which we shall afterwards make an interesting application.

From (1), replacing x, y by $1, x$ and taking $n = 0, 1, 2, 3 \dots$, the formulas

$$\begin{aligned} 1 &= B_0^{(0)}, \\ 1 + x &= B_0^{(1)} + B_1^{(1)}x, \\ (1 + x)^2 &= B_0^{(2)} + B_1^{(2)}x + B_2^{(2)}x^2, \\ &\dots \\ (1 + x)^n &= B_0^{(n)} + B_1^{(n)}x + B_2^{(n)}x^2 + \dots + B_n^{(n)}x^n \end{aligned} \quad (2)$$

are obtained.

We now use the well-known summation formula for the geometric series:

$$1 + (1 + x) + (1 + x)^2 + \dots + (1 + x)^n = \frac{(1 + x)^{n+1} - 1}{x},$$

where, if $(1 + x)^{n+1}$ is again expanded by the binomial formula, replacing n by $n + 1$ in the last formula (2) and setting $B_0^{(n+1)} = 1$,

$$\frac{(1 + x)^{n+1} - 1}{x} = B_1^{(n+1)} + B_2^{(n+1)}x + \dots + B_{n+1}^{(n+1)}x^n. \quad (3)$$

Comparing this with the sum of the right hand sides of (2), and, according to the rules of § 1, setting equal the coefficients of corresponding powers of x , one obtains

$$\begin{aligned} B_0^{(0)} + B_0^{(1)} + B_0^{(2)} + \cdots + B_0^{(n)} &= B_1^{(n+1)}, \\ B_1^{(1)} + B_1^{(2)} + \cdots + B_1^{(n)} &= B_2^{(n+1)}, \\ &\dots \\ B_n^{(n)} &= B_{n+1}^{(n+1)}, \end{aligned} \tag{4}$$

or in general

$$B_\nu^{(\nu)} + B_\nu^{(\nu+1)} + \cdots + B_\nu^{(n)} = B_{\nu+1}^{(n+1)}. \tag{5}$$

If however the equations (2) are multiplied in order by

$$B_0^{(n)}, \quad -B_1^{(n)}, \quad +B_2^{(n)}, \dots, \quad \pm B_n^{(n)},$$

where the upper sign is to be taken for even n and the lower for odd n , the sum of the left hand sides gives

$$B_0^{(n)} - B_1^{(n)}(1+x) + B_2^{(n)}(1+x)^2 - \cdots \pm B_n^{(n)}(1+x)^n = [1 - (1+x)]^n = (-x)^n,$$

and equating the coefficients of equal powers on the right and left hand sides yields the system of formulas

$$\begin{aligned} 0 &= B_0^{(0)} B_0^{(n)} - B_0^{(1)} B_1^{(n)} + B_0^{(2)} B_2^{(n)} - \cdots \pm B_0^{(n)} B_n^{(n)}, \\ 0 &= -B_1^{(1)} B_1^{(n)} + B_1^{(2)} B_2^{(n)} - \cdots \pm B_1^{(n)} B_n^{(n)}, \\ &\dots \\ \pm 1 &= \pm B_n^{(n)} B_n^{(n)}, \end{aligned} \tag{6}$$

and this system of formulas can also be written in the reversed order, using the relation $B_\nu^{(n)} = B_{n-\nu}^{(n)}$, as

$$\begin{aligned} 1 &= B_0^{(n)} B_0^{(n)}, \\ 0 &= B_1^{(n)} B_0^{(n)} - B_0^{(n-1)} B_1^{(n)}, \\ 0 &= B_2^{(n)} B_0^{(n)} - B_1^{(n-1)} B_1^{(n)} + B_0^{(n-2)} B_2^{(n)}, \\ &\dots \\ 0 &= B_n^{(n)} B_0^{(n)} - B_{n-1}^{(n-1)} B_1^{(n)} + B_{n-2}^{(n-2)} B_2^{(n)} - \cdots \pm B_0^{(0)} B_n^{(n)}, \end{aligned} \tag{7}$$

or, in a summarizing notation,

$$\sum_{\nu=0}^{\mu} (-1)^\nu B_{\mu-\nu}^{(n-\nu)} B_\nu^{(n)} = \begin{cases} 1, & \mu = 0, \\ 0, & \mu = 1, 2, \dots, n. \end{cases} \tag{8}$$